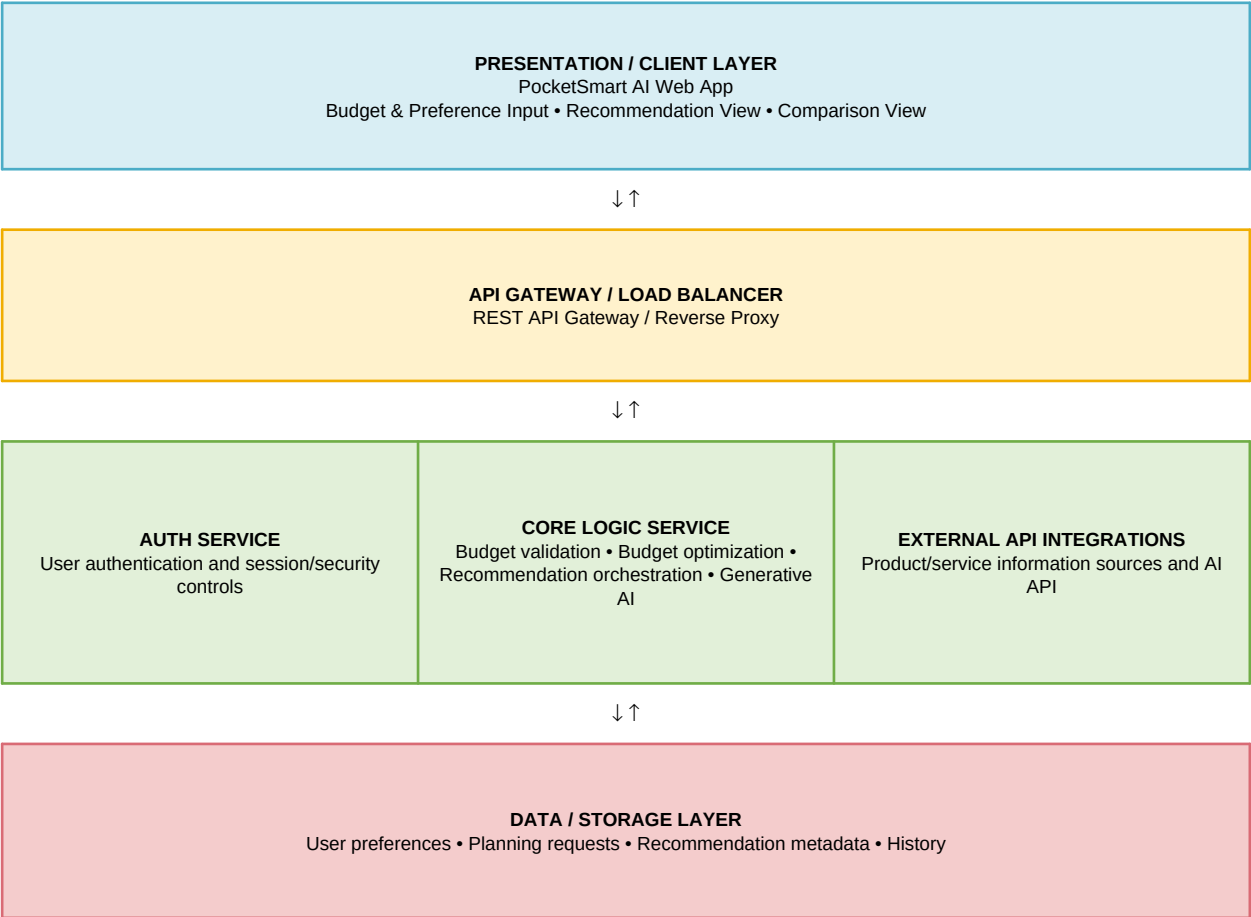


Solution Architecture

Date	15 March 2026
Team ID	xxxxxx
Project Name	PocketSmart AI
Maximum Marks	5 Marks

Solution Architecture Diagram

High-level architecture for PocketSmart AI. The flow follows the template's Presentation / Client Layer → API Gateway → Application Services → Data / Storage Layer structure.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects budget, category and preferences and presents personalized recommendations, budget allocation and comparison results.	React, HTML/CSS, JavaScript
API Gateway	Provides the entry point for client requests, routing them to the appropriate application service and supporting controlled API access.	REST API, reverse proxy / gateway
Microservices / Core Logic	Validates inputs, applies budget rules, orchestrates AI recommendations, ranks results and prepares comparison data.	Python, FastAPI, recommendation/AI service
Database	Stores planning requests, preferences, recommendation metadata and relevant history needed for the application.	PostgreSQL
External Integrations	Supplies AI capabilities and product/service information used to create and enrich recommendations.	Generative AI API; external product/service APIs